

Structural Biases in LLM-as-a-Judge Systems: Implications for reliable IS Adoption

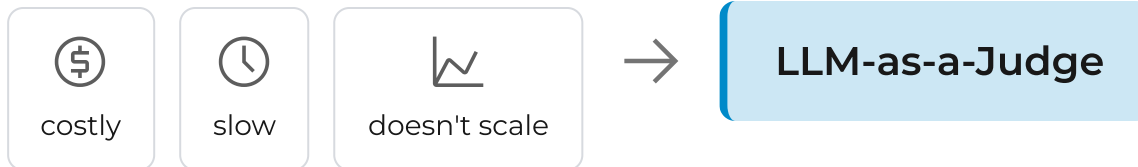
Can we trust AI to judge content?

Cédric Gaspoz, Ulysse Rosselet et Jonathan Piaget

August 22, 2026

What is LLM-as-a-Judge

human evaluation



already deployed today



Pre-screening applications



Assisting juries



Code review

Issues

- sensitivity to prompts and language
- variability depending on the task
- the existence of systematic biases, such as:

Self-preference bias

the judge favors the content it produced itself, or that of closely related models

AI-AI bias

the judge prefers machine-written content over human-written content

Position bias

the order in which the two candidates are shown changes the verdict

... and others

One number to keep in mind:

23–60%

more likely to be **shortlisted** when the candidate used
the **same LLM as the recruiter**

Simulated hiring pipeline - Xu, J. et al., 2025

Dictionaries as a Human Gold Standard

On subjective benchmarks, human answers vary too much to separate **judge bias** from **content quality**.

Professional

curated reference sources with
verified non-LLM provenance

Comparable

same task, same target format, so
sources sit side by side

Accessible

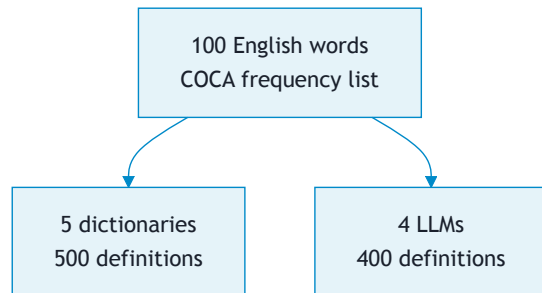
freely available entries

If strong biases appear even in this narrow, controlled setting, there is no reason to expect them to disappear in fuzzy organizational tasks.

Research Question

To what extent do intrinsic biases - such as AI-AI bias, self-preference and position - pose risks to reliable acceptance of the LLM-as-a-Judge paradigm in organizational context?

Corpus



Generation prompt - deliberately minimal:

"You are a dictionary editor. For each word received, produce a dictionary definition entry. Include: part(s) of speech, word senses with definitions, and usage examples where helpful."

Dictionaries: Cambridge, Collins, Merriam-Webster, Oxford ALD, Wiktionary

LLMs: DeepSeek R1, GPT-5, Llama 3 8B (local), Mistral Large

Evaluation Protocol

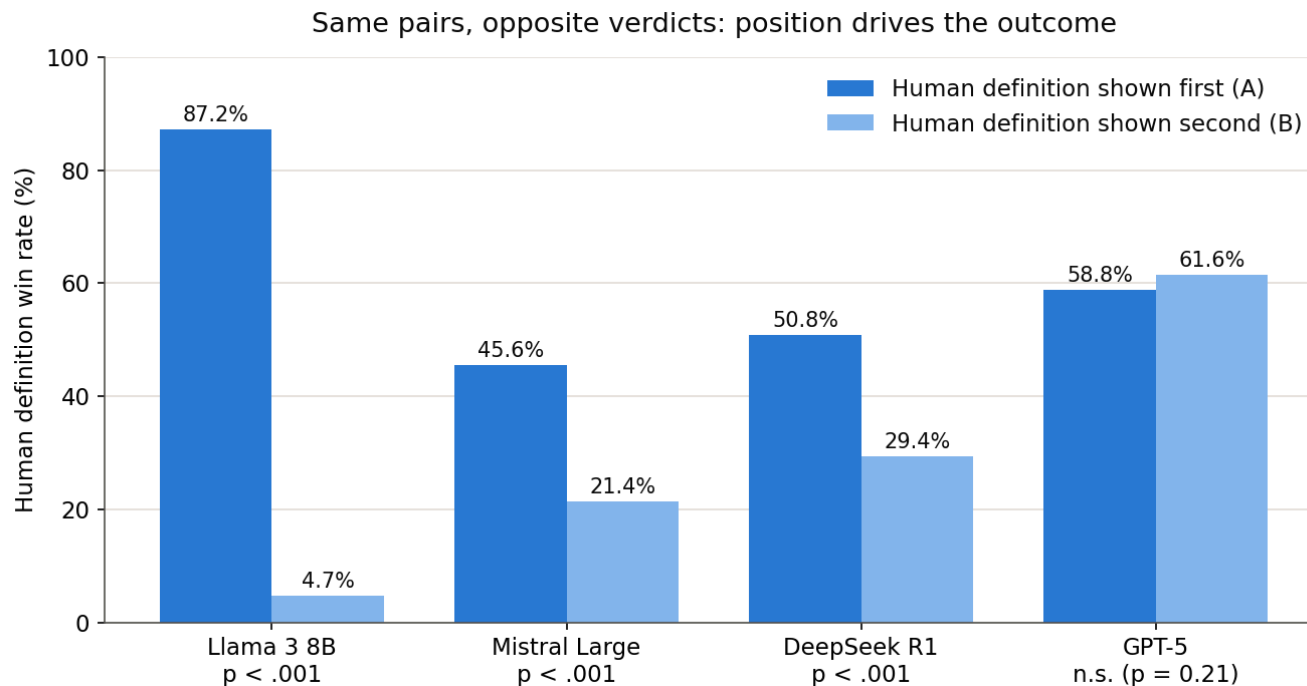
- **Blind pairwise comparison:** definition A vs definition B
- **Anonymized:** the judge never knows who wrote what
- **Randomized order:** A first or B first, 50/50

4 LLMs × 5 dicts	→	20 pairs/word
20 × 100 words	→	2,000 pairs
2,000 × 4 judges	→	8,000 judgments

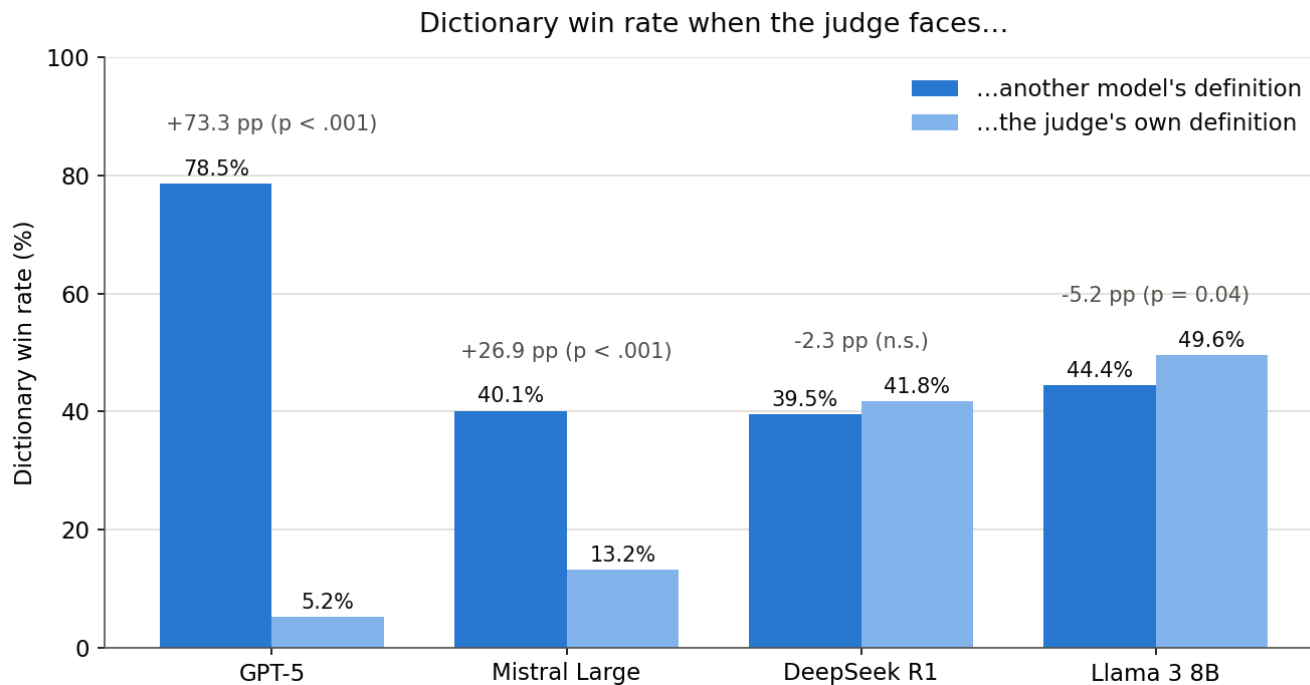
Judge prompt - role, no criteria:

*"You are the editor of an English language dictionary.
Two writers have proposed definitions for a given word.
It is up to you to decide which definition to include in
the dictionary."*

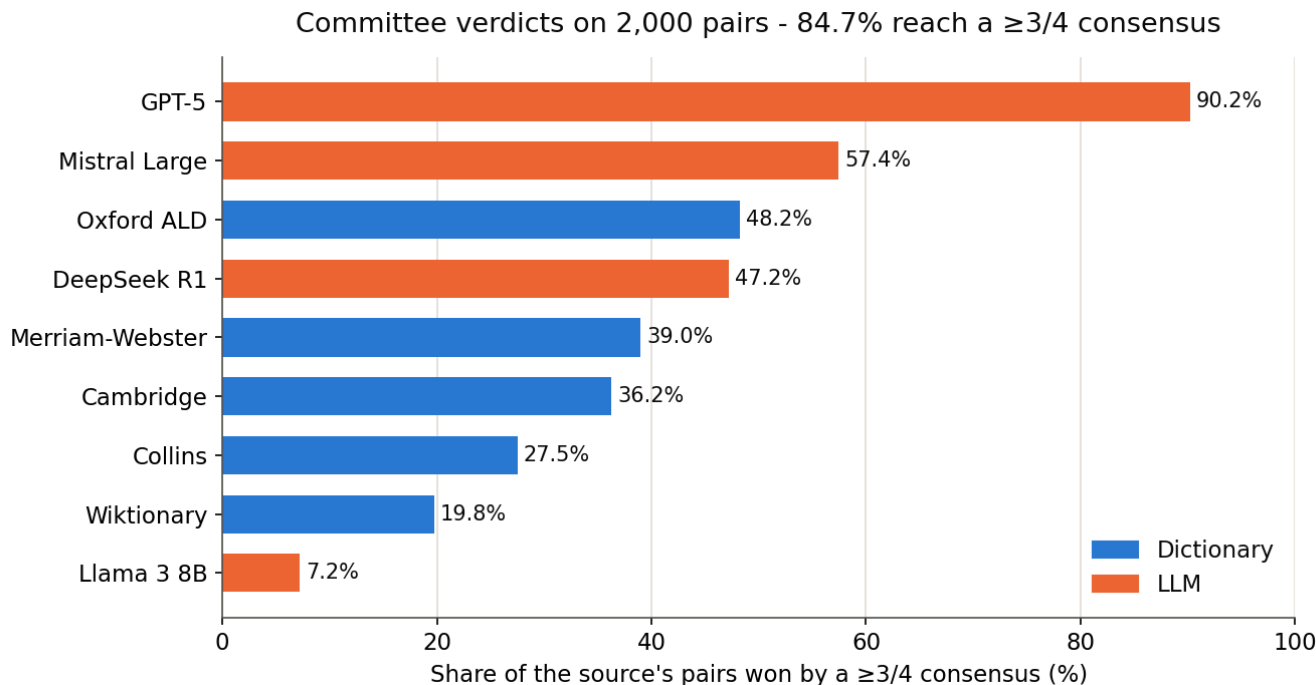
Result 1 - Position Bias: 3 of 4 Judges



Result 2 - Self-Preference Bias

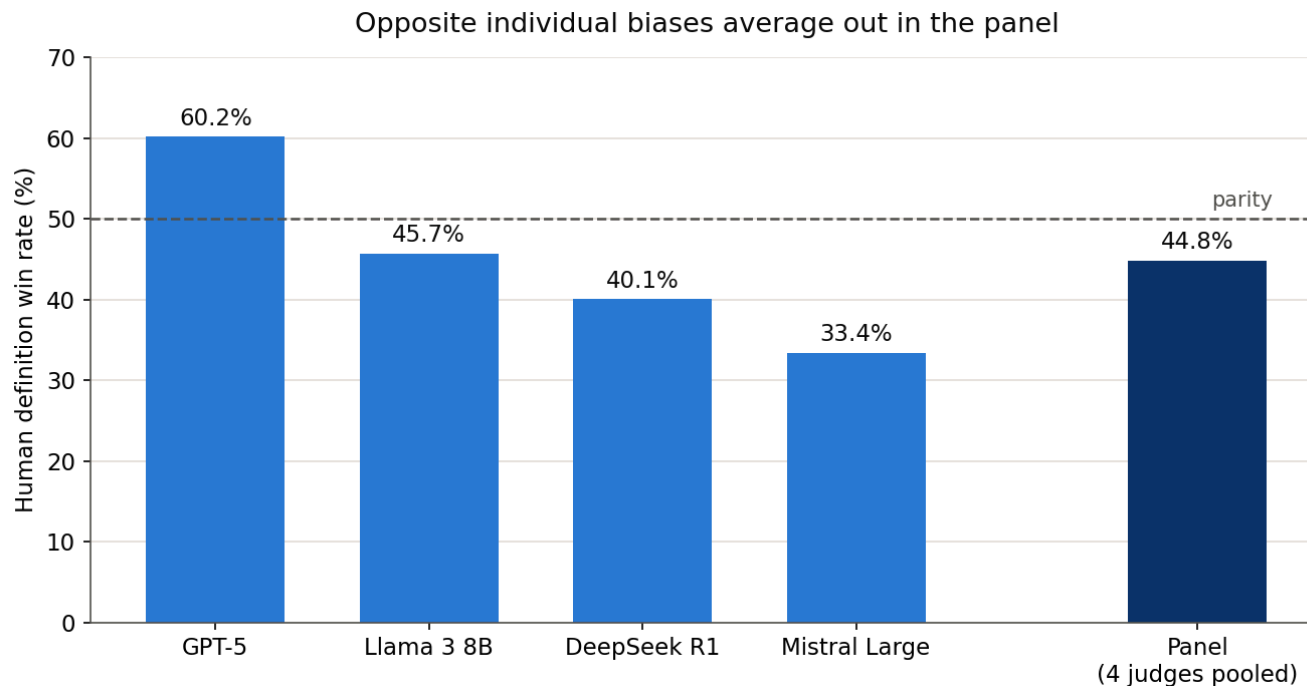


Result 3 - Consensus Analysis



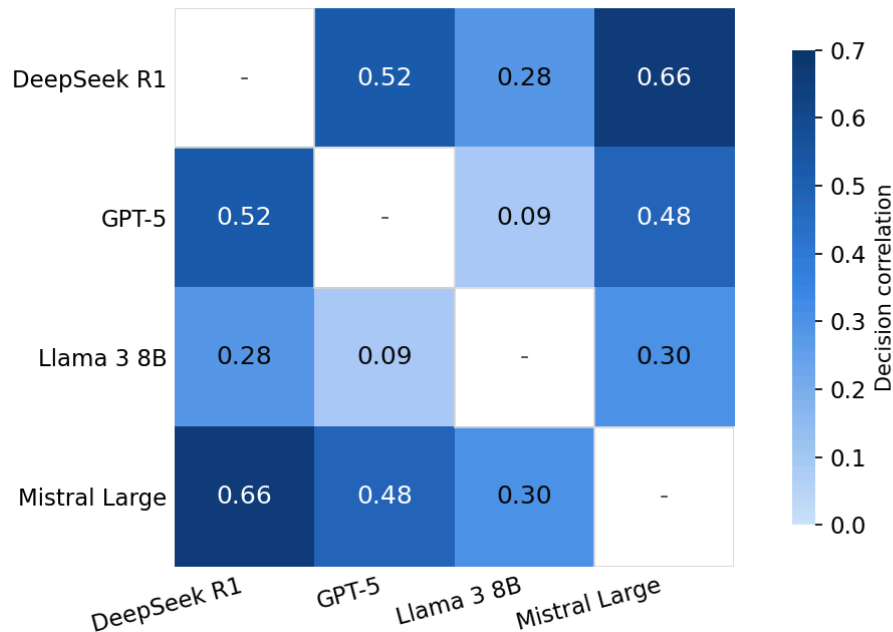
Committee level: LLM definitions win 50.5% of pairs, dictionaries 34.2%, no consensus 15.3%

Result 4 - Win Rates & the Panel Effect



Result 5 - Inter-Judge Agreement

Judges barely agree with each other - Fleiss $\kappa = 0.357$ (moderate)



Limitations

- **Scope:** 100 high-frequency English words, one genre (lexicographic definitions)
- **No human control group:** we measure disagreement between judges, not accuracy
- **One prompt, one run per pair:** prompt sensitivity and self-consistency untested
- **4 judges:** enough to show the effect, too few to model it
- **Consensus $\neq$ ground truth:** converging judges can converge on the wrong answer

These limits bound the **magnitude** of our numbers - not the **existence** of the biases. Position and self-preference are measured on identical content.

Conclusion

Narrow, controlled setting

curated human references, blind protocol

Biases still massive

- position - first shown wins **65.7%** (3/4 judges)
- self-preference - up to **+73.3 pp**
- AI-AI - **55.2%** for synthetic text

Sophistication $\neq$ impartiality → a judge not perceived as neutral will not be adopted

 **Measure** the biases of your judges

 Deploy **blind, diverse panels**

 **Calibrate continuously** against human ground truth

Thank You!

Questions?

✉ jonathan.piaget@he-arc.ch

🌐 piagetjonathan.ch

